TANISHK GANGWAR

B.Tech CSE (Data Science) · Manipal University Jaipur · Batch 2028

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CS junior with hands-on experience in causal inference, statistical modeling, and production ML — built and deployed 3 end-to-end systems, including a quasi-experimental study across 403K commits quantifying how GitHub Copilot changes developer commit behavior. Targeting SWE, ML, and DS internships.

EDUCATION

B.Tech - Computer Science & Engineering (Data Science), Manipal University Jaipur Oct 2024 – Present

CGPA: 9.00 / 10 - Dean's Academic Excellence Award, three consecutive semesters (Sem 2, 3 & 4)

Coursework: Data Structures & Algorithms · Machine Learning · Data Mining · DBMS · OOP · Computer Networks · Cloud Computing · Operating Systems

TECHNICAL SKILLS

Languages	Python, C++, Java, SQL, JavaScript
Backend / APIs	REST APIs, FastAPI, Flask, Docker, Git Workflows, Async Python
ML / Data	PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, CLIP, UMAP, XGBoost, Feature Engineering, Data Visualization
Statistics	Causal Inference (DiD), Statistical Modeling, Hypothesis Testing, A/B Testing, Cross-Validation, Information Theory
Tools	Git, GitHub, VS Code, Jupyter Notebook, Railway (Cloud Deployment), MLOps, Linux/Bash

PROJECTS

Sekivara - Causal Analysis of GitHub Copilot's Behavioral Fingerprint GitHub Python · Pandas · Statsmodels · DiD

- Architected a large-scale **data pipeline** ingesting **~403,000 commits** across **9 open-source repos** (2018–2024) via automated GitHub API ingestion; designed a quasi-experimental **difference-in-differences** framework (HC3 robust SE, parallel-trends validated); Copilot adoption causally reduced mean files/commit by **28%**, insertions/commit by **37%**, and large-commit fraction by **2.4pp** (all $p < 0.01$), driven entirely by returning contributors across **2 cohorts and 3 post-treatment periods**.

Entropic Fingerprint - Shannon Entropy Release Detection (Negative Result) GitHub Python · Scikit-learn · Information Theory

- Developed and validated a **5-feature Random Forest pipeline** (commit cadence, entropy delta, author diversity, churn rate, burst score) over **6 years** of history (Flask, Django, CPython); initial **72% accuracy** collapsed to **AUC 0.47** under leave-one-repo-out CV; diagnosed confound (**Spearman r = 0.817, p = 0.007** between commit volume and entropy); published as a rigorous negative result with open-sourced replication code.

Kansei - CLIP-Powered Aesthetic Identity Engine GitHub Python · FastAPI · CLIP · UMAP · Deployed

- Architected and deployed a production end-to-end ML system encoding visual preferences into CLIP's **512-dimensional embedding space** via **25 forced-choice comparisons**, scored against **16 aesthetic centroids** using cosine similarity; served via a production FastAPI backend with async startup preheating at kansei.up.railway.app.
- Engineered a **4-component inference pipeline** (weighted preference vectors, kNN over **100+ embeddings**, rejection analysis, pairwise consistency scoring) visualized via **live 3D UMAP**; parallelized **17 simultaneous API calls** at quiz init with preloaded **16 centroid vectors**, reducing perceived load to a **single round-trip**.

ACHIEVEMENTS & AWARDS

- **Google - The Big Code** - ranked in the **Top 1,500 nationally** among thousands of participants in Google's national competitive programming event; Top 1,500 out of 100,000+ registered participants.
- **Smart India Hackathon (SIH) 2025** - College Winner; selected among the winning teams at the college level.